

Loc Nguyen

Email: locnguyen16022003@gmail.com | 0474202372 | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Data Analyst with industrial experience delivering end-to-end analytical projects across predictive modelling, customer segmentation, and automated data pipelines on large-scale datasets. Proficient in SQL, Python, and R across the full data lifecycle, converting ambiguous business problems into structured analytical approaches that directly inform strategic decision-making. Experienced in building interactive Power BI dashboards to communicate findings clearly to senior stakeholders across operational, and product functions. Delivered measurable outcomes including 94% traffic-violation detection precision, 99% anomaly detection accuracy, 20% misclassification reduction, and 30% lower manual pipeline intervention. Curious, proactive, detail-oriented analyst who communicates impactful data stories clearly in fast-paced environments.

EXPERIENCE

ONETASK

Data Analytics Intern

Melbourne, VIC

July 2025 - November 2025

- Developed and benchmarked **YOLOv4**, **YOLO-NAS**, and **RF-DETR** computer vision models for traffic violation detection on Linux systems, producing proof-of-concept evaluations that informed model selection and business recommendations for future production deployments.
- Operationalised detection models enhanced with the Red-Fox Optimisation (RFO) algorithm, achieving **94% mAP** for seatbelt detection and **82% mAP** for phone-use detection, moving solutions from experimentation to reliable, real-world traffic safety applications.
- Built an Apache Airflow pipeline to automate end-to-end training, testing, inference, and visualisation workflows within an agile delivery methodology, reducing manual intervention by **30%** and improving analytical throughput and reproducibility.
- Designed interactive model-performance visualisations to communicate trade-offs across experiment settings (worker count, model layers, learning rate) to operational managers and senior stakeholders, translating technical findings into strategic recommendations.

CONSTELLATION TECHNOLOGIES LIMITED

Data Analytics Intern

Melbourne, VIC

July 2024 - November 2024

- Developed a **BERT-based NLP question-answering system** to extract structured data from unstructured food product records, achieving **95%** extraction accuracy and improving the speed and consistency of data retrieval.
- Built an **Isolation Forest** anomaly detection model to identify abnormal temperature patterns with **99% accuracy**, supporting earlier detection of quality risks in agricultural supply chains and maintaining products in excellent condition.
- Produced interactive **Power BI** dashboards to visualise temperature trends, key operational drivers, and automated alerts for deviations beyond recommended storage thresholds, enabling senior stakeholders to monitor performance and make faster, data-driven supply chain decisions.

ONETASK

Data Evaluation Specialist

Melbourne, VIC

March 2023 - April 2026

- Reviewed up to **3,500** traffic enforcement images daily, producing true-positive and false-negative statistics that directly supported computer vision **model evaluation** and operational reporting for road safety systems.
- Conducted monthly data quality audits using Excel (VLOOKUP, pivot tables) to validate outcomes against verified records, reducing the **misclassification rate** by **20%** and improving the reliability of labelled data used for strengthening model accuracy and strategic planning.
- Created **Power BI visualisations** of offence trends by highway and camera location, communicating high-volume hotspot patterns of **300+** daily offences clearly to stakeholders and supporting data-driven enforcement prioritisation.

- Identified systemic image quality issues (blur, poor lighting) and escalated to engineering teams, enabling faster camera recalibration and preventing downstream data loss in road safety monitoring pipelines.

EDUCATION

Monash University

Bachelor of Applied Data Analytics Advanced (Honours)

Grade: First Class Honours (H1)

Honours Thesis: Application of Detection Transformers in Live Traffic Monitoring Systems

Dean's List 2023

Clayton, Australia

December 2025

PROJECTS

EUROPEAN BANK CHURN ANALYSIS

- Extracted and queried large customer datasets using **SQL** to build **RFM segmentation** and **Power BI dashboards** surfacing retention risks and churn patterns; managed experiment tracking with **MLflow** and hyperparameter optimisation with **Optuna** to select the best **XGBoost** model (**72% overall accuracy**).
- Deployed churn insights via a **Streamlit** interface, translating complex model outputs into accessible, decision-ready visuals.

ROSSMANN FORECASTING PROJECT

- Led a five-member team through **full data lifecycle** delivery: preprocessing and feature engineering in Python (Pandas), KNN customer sales segmentation (**98% accuracy**) and Random Forest forecasting (**95% accuracy** in 4-week sales prediction).

BUREAU OF METEOROLOGY PROJECT

- Collaborated with a seven-member team to clean, resample, and interpolate raw meteorological datasets; compared KNN, Logistic Regression, and Random Forest models, selecting Random Forest as the best performer (**83% classification accuracy**).

TECHNICAL SKILLS

Languages: SQL, Python, R

ML & Modelling: scikit-learn, PyTorch, XGBoost, Random Forest, BERT, YOLO (v4/NAS), RF-DETR, Isolation Forest

Data & Pipelines: Pandas, Apache Airflow, MLflow, Optuna, OpenCV, Docker

Visualisation & BI: Power BI, Matplotlib, Streamlit

Infrastructure: Git, Linux (Ubuntu), Docker

Core Competencies: Predictive Modelling, Customer Segmentation, Full Data Lifecycle, Agile Delivery, Stakeholder Communication, Business Performance Reporting,